ASTANA IT UNIVERSITY · 2026

Pre-defense 1 · Report

AI-Based Platform to Support the Employment of Astana IT University Graduates

24 October 2026

> Revised after supervisor feedback of 24 October 2026. AI scope, evaluation protocol and contribution evidence are clarified. Independent matching evaluation and GitHub peer review remain next-stage actions.

Team size	2
Student 1 — full name	Қожабек Аяділ Нұрланұлы
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Supervisor position	Senior Lecturer
Repository URL	https://github.com/vbl0-0/AITU-Career
Issue tracker URL	https://github.com/users/vbl0-0/projects/1

Submission deadline: 24 October 2026 before 18:00 (UTC+5).

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1. Project Proposal and Scope

AI-Based Platform to Support the Employment of Astana IT University Graduates.

Problem statement. Graduates need to compare their educational and practical experience with heterogeneous IT job requirements and identify an actionable next step. The existence and scale of this need specifically among AITU graduates remain hypotheses until local user discussions are completed. Research on skill extraction supports the technical relevance of the task, but does not establish local demand [S01–S05].

Target users are senior AITU students and graduates, plus an administrator of the educational prototype. A career-center representative is a potential requirements expert. An employer dashboard is outside the current MVP.

Aim. Develop and evaluate a reproducible prototype that explains how a graduate profile covers IT vacancy skill requirements, identifies skill gaps and suggests development actions. The matching percentage is not a probability of employment.

Objective	Verifiable outcome
Investigate the domain	At least 12 meaningfully used sources including eight scientific publications; comparison of four systems. Optional user research is planned for later validation.
Specify requirements	A priority, acceptance criterion and verification method for each mandatory requirement.
Design the system	System context, use cases, components, data relationships and API responsibilities.
Implement the core journey	Profile → skills → vacancy → explained match → development actions.
Evaluate the approach	Compare exact matching, normalization and optional semantics under one fixed protocol.
Prepare submission evidence	Unified report, editable diagrams, traceable tasks and actual verification evidence.

Included: authentication and graduate/admin roles; AITU-oriented profiles and education; skills with proficiency levels; projects, certificates and experience; vacancies and saved jobs; transparent weighted matching; saved analyses with staleness detection; recommendations and administration. The local seed contains 30 fictional vacancies clearly identified as demonstration data.

Excluded from this stage: automated hiring decisions, validated prediction of employment, university SSO/DU/Moodle integration, large-scale collection of external vacancies, production SLA, employer accounts and training a large language model. Visual similarity to university services does not imply integration or institutional endorsement.

The team consists of two students. Confirmed names and GitHub accounts are recorded below. Actual contributions are recorded when supported by completed work and review evidence.

Team member	Full name	GitHub account	Agreed contribution
1	Қожабек Аяділ Нұрланұлы	https://github.com/vbl0-0	Backend/API and migration delivery, backend regression tests and repository automation; publication and verification assisted by OpenAI Codex. Evidence: https://github.com/vbl0-0/AITU-Career/pull/19 ; https://github.com/vbl0-0/AITU-Career/pull/21 ; commits 6c434f2 and 0b8c8bb. These records describe current delivery, not exclusive initial authorship of all imported code.
2	Амантай Жан Әділұлы	https://github.com/igiiaw	Frontend, multilingual interface, research documentation and report-source delivery; publication and verification assisted by OpenAI Codex. Evidence: https://github.com/vbl0-0/AITU-Career/pull/20 ; https://github.com/vbl0-0/AITU-Career/pull/22 ; commits 7717652, c12021f and 203f7c7. These records describe current delivery, not exclusive initial authorship of all imported code.

2. Evidence of Relevance and Optional User Research

On 24 October 2026, the team clarified that user interviews and supervisor feedback are optional for this submission. This is a team-reported requirement clarification, not a recorded supervisor approval.

Relevance at this stage is supported by the literature and system comparison in Sections 3–4. No user interviews were conducted and no interview results are presented. Local demand, explanation usefulness and the employment-readiness indicator remain hypotheses rather than empirically established conclusions.

Claim	Available evidence	Limit
Skill extraction and normalization require explicit design decisions.	Scientific publications S01–S05.	Transfer to Kazakh/Russian and AITU profiles needs local evaluation.
Existing services combine vacancy matching and skill development.	Official system materials S11–S14 and the common-criteria comparison.	This does not prove unmet local demand or superiority of this prototype.
Explanations can expose covered, partial and missing skills.	Implemented calculation and automated functional checks.	User comprehension and usefulness have not been measured.
Employment readiness summarizes profile and vacancy evidence.	Documented configurable heuristic.	It is not a validated predictor of hiring success.

Future optional research may include anonymized discussions with graduates, career-center staff and recruiters. Findings should be recorded with actual dates and consent, separated from interpretations and linked to requirement decisions. Such evidence is not a prerequisite for this submission and no participant identities or fabricated responses are included.

3. Critical Literature Review

A targeted engineering search was performed on 19 October 2026 using ACL Anthology, arXiv and official product/technical documentation. Queries covered skill extraction from job postings, multilingual skill matching, sentence embeddings and job-title representations. Inclusion required verifiable metadata and a direct connection to extraction, retrieval or an engineering decision. This is not claimed to be an exhaustive systematic review.

The review uses 17 sources: eight peer-reviewed scientific publications S01–S08, technical report S09 and eight official materials S10–S17. The scientific core is from 2021–2025; Sentence-BERT (2019) is retained as foundational work. The limitations and applications below are the project's analysis of transferability, not verbatim author claims.

Source and approach	Limitation for this project	Engineering implication
S01. Expert annotation of hard and soft skill mentions and domain-adapted extraction models.	English job postings do not establish performance on Russian/Kazakh texts or AITU profiles.	Evaluate extraction separately from ranking; retain reference spans and annotation rules.
S02. Survey of terminology, datasets and neural skill extraction methods.	Different datasets and task definitions prevent direct comparison of reported scores.	Define the annotation unit, skill terminology and normalization rules before evaluation.
S03. Multilingual domain-adaptive pre-training using the ESCO taxonomy.	Transfer to local occupations and languages requires validation; large-model adaptation needs resources.	Maintain canonical skills and aliases; consider ESCO alignment as a future extension.
S04. Few-shot LLM extraction explores complex skill mentions.	The study does not establish universal superiority over supervised methods; inference requires quality control and resources.	Keep external LLM calls optional and assess extraction against fixed labels.
S05. Multilingual vacancy–candidate matching through skill extraction, ESCO alignment and a knowledge graph.	The graph pipeline is more complex than a dictionary, and its benefit for AITU graduates remains unmeasured.	Compare normalization with semantic expansion and retain the origin of each match.
S06. Job-title representations incorporate information from descriptions beyond extracted skills.	Learning job-title representations is different from assessing an individual candidate.	Describe the current score as skill-requirement coverage, not a probability of being hired.
S07. Siamese sentence encoders enable efficient embedding-based text comparison.	Text similarity does not prove skill proficiency or satisfaction of all job conditions.	Use similarity only as supplementary evidence with a limited contribution.
S08. Contrastive learning produces sentence representations in supervised and unsupervised settings.	Semantic textual similarity benchmarks do not replace domain-specific matching evaluation.	Select an encoder using local ranking quality and latency as well as general benchmarks.
S09. A family of multilingual retrieval embeddings offers different model sizes.	Input conventions, model size and inference cost affect suitability.	Include E5 in a future comparison; it is not the model currently configured in the prototype.

Synthesis and research questions

S01–S04 motivate separating a skill mention, its canonical label and evidence of proficiency. The MVP therefore distinguishes an explicitly reported skill/level from inferred text evidence. Dictionary matching is reproducible but limited by catalog coverage and language variation. S03 and S05 motivate taxonomy alignment without establishing a need for a graph database at this scale.

S06 highlights information in vacancy descriptions beyond skill lists. The current score consequently measures only coverage of explicit or extracted skill requirements. S07–S09 justify evaluating embeddings against literal matching, but textual proximity alone does not establish competence. Semantic evidence is therefore capped at a partial contribution and its method is recorded.

RQ1: do alias normalization and optional semantic matching improve nDCG@5 over exact skill matching on a fixed profile/vacancy dataset while retaining interpretable explanations? RQ2: do users distinguish skill coverage from the probability of employment? The contribution is a testable engineering adaptation to AITU. Superiority over commercial systems is not asserted.

4. Existing System Comparison

Four relevant systems are compared on eight consistent criteria using official public documentation: LinkedIn [S11], Handshake [S12], hh.kz [S13] and Coursera Career Academy [S14]. No private product accounts were audited. “Not established” means that the cited sources do not establish the capability; it does not mean the capability is absent.

Criterion	LinkedIn	Handshake	hh.kz	Career Academy
Primary purpose	Job discovery and matching	Early-career discovery and recommendations	Regional job search	Role-oriented skill development
User information	Profile and resume	Profile, interests and activity	Resume and key skills	Learning pathway; vacancy matching not established
Personalized vacancies	Documented	Documented	Exact mechanism not established by S13	Not established
Skills and requirements	Profile and job qualifications	S12 emphasizes interests and activity	Key skills	Skills associated with a career role
Match explanation	Documented	Per-requirement explanation not established	Not established	Role skills rather than individual vacancy fit
Development actions	Profile improvement and upskilling	Career recommendations; detail not established	Not established	Courses and practical projects
Published exact scoring formula	Not established	Not established	Not established	Not established
AITU-program alignment	Not established	Not established	Not established	Not established

Matching and skill-development recommendations already exist in competing systems. The rationale for this prototype is controllable AITU-specific adaptation, an accessible per-requirement calculation and reproducible evaluation on a fixed dataset. Optional future user research can test the local-demand hypothesis. Commercial systems are not quantitative baselines because their internal algorithms and evaluation data are unavailable.

5. Technology Comparison and Architectural Decisions

The alternatives are assessed for a small educational prototype with a Python analysis module, relational data and no interaction history. Assessments are qualitative; no unperformed performance measurements are implied.

Decision	Reasonable alternatives and trade-offs	Selected solution and rationale
Backend platform	FastAPI: validation/OpenAPI and direct Python NLP integration [S15]. Django REST: mature conventions and administration, with a larger framework surface. Node.js/NestJS: a shared frontend language, but Python NLP needs an additional integration boundary.	FastAPI supports one small API/analysis process and a modular code structure.
Data storage	PostgreSQL: relational constraints and transactions [S16]. SQLite: simple setup with a different concurrency/operation profile. MongoDB: flexible documents, with a different strategy for cross-record integrity.	PostgreSQL with SQLAlchemy and Alembic preserves linked profiles, catalog entries and vacancy requirements.
Matching approach	Exact matching: transparent but sensitive to wording. Dictionary/aliases/weights: explainable but requires catalog maintenance. Embedding hybrid: supports paraphrases [S07–S09] but adds calibration and inference cost.	Weighted normalized keyword matching is the default; embeddings remain optional and require separate evaluation.
Component communication	REST/JSON: straightforward CRUD and OpenAPI. GraphQL: flexible selections with an additional schema/cost-control layer. gRPC: typed binary communication but more integration work for a browser client.	REST/JSON under /api, with Vite/nginx proxying to a single browser origin.
Frontend	React/TypeScript: typed components already implemented. Vue/TypeScript: a comparable SPA option requiring component migration. Server templates: a smaller client stack but a different approach to interactive profile editing.	React/TypeScript with Vite follows the implemented journeys; no unmeasured speed advantage is claimed.

ADR-01: use a modular monolith. Authorization, API, analysis and persistence have separate code responsibilities but share one backend deployment. Revisit separation only if profiling demonstrates interference from inference. ADR-02: PostgreSQL is the source of truth; preserve algorithm version and input revisions with saved analyses. ADR-03: absence of the optional model must not break keyword analysis.

6. Requirements Specification and Acceptance Criteria

MoSCoW priorities: Must denotes mandatory MVP behavior; Should denotes a planned improvement; Could denotes an optional extension. An implemented feature is not evidence that users need it. Requirement IDs connect the specification, components, tasks and verification.

ID / priority	Requirement	Acceptance criterion	Verification
FR-01 / Must	Registration and sessions	A new email creates a graduate account; invalid passwords are rejected; logout invalidates the previous access token.	API authentication tests; browser login/logout
FR-02 / Must	Authorization boundaries	Other users' records are inaccessible; graduates cannot execute admin actions or grant themselves an admin role.	API ownership/role tests
FR-03 / Must	Profile and education	Program, graduation year and desired role are saved; an invalid graduation year is rejected.	API validation; graduate E2E
FR-04 / Must	Skills and proficiency	Catalog skills use levels 1–3; duplicates and unknown IDs are rejected.	API skill tests
FR-05 / Must	Portfolio	Projects, technologies, certificates and experience support CRUD; an end date before the start date is rejected.	API and portfolio E2E
FR-06 / Must	Vacancies and saved jobs	Search/filters return matching records; details expose requirements; repeated save/remove operations are handled consistently.	API vacancy tests; graduate E2E
FR-07 / Must	Deterministic matching	Identical keyword inputs/configuration produce the same score; four of five equal full matches yield 80%; no requirements yield null.	Unit matching tests
FR-08 / Must	Explanations	Each requirement belongs to matched/partial/missing; contribution, weight, method, version and gap-based actions are available.	Matching tests; detail/analysis inspection
FR-09 / Must	Stale analysis	Profile, vacancy or relevant context changes flag saved results as stale and offer recalculation.	API fingerprint/staleness tests; E2E
FR-10 / Must	Administration	Admins manage jobs/catalogs; deleting a referenced category is handled without data loss.	API constraints; admin E2E
FR-11 / Should	Semantic extension	No model triggers keyword fallback; a semantic match is labeled and contributes at most 0.5.	Controlled semantic/fallback tests; real-model evaluation separately
FR-12 / Must	Demonstration-data labels	Fictional vacancies are labeled; readiness reports its sample scope and components.	UI inspection; career summary

ID / priority	Quality attribute	Acceptance criterion	Verification
NFR-01 / Must	Security	Argon2id passwords, signed JWTs, HttpOnly refresh cookies, hashed refresh secrets and rejection of untrusted origins.	API tests; configuration review
NFR-02 / Must	Integrity	Foreign keys, UNIQUE/CHECK constraints and migrations are enforced; repeat seeding creates no duplicates.	PostgreSQL tests; Alembic check
NFR-03 / Must	Reproducible startup	A supported clean environment can run setup, migrations, seed and health checks; native Windows and Compose instructions are separate.	Runbook; container startup not yet verified
NFR-04 / Must	Mobile usability	The core journey fits 390×844 without horizontal overflow; menu, inputs and actions remain available.	Playwright and screenshots
NFR-05 / Should	Accessibility	Keyboard focus is visible; normal text contrast is at least 4.5:1; errors have textual descriptions.	Targeted audit; comprehensive audit remains outstanding
NFR-06 / Should	Latency	Target: keyword matching p95 ≤500 ms on 30 demonstration vacancies after warm-up; no measurement is claimed.	30 warm-up and 200 measured requests on fixed hardware
NFR-07 / Must	Traceability	Saved results include algorithm version, mode, revisions and a context fingerprint.	Unit/API fingerprint tests
NFR-08 / Must	Confidentiality	Submission files exclude .env/tokens and identifiable interview data.	Package inventory and staged-diff review

Matching and readiness rules

Score = 100 × Σ(weight × contribution) / Σ(weight). For an explicit skill, contribution = min(1, user_level / required_level); for an absent skill it is zero. Inferred text and semantic evidence are capped at 0.5. Required/preferred skill weights default to 2/1 and may be set to other positive values. Scores across different vacancies do not imply equal job difficulty.

Readiness is an internal heuristic: market 60%, projects 15%, experience 10%, certificates 5% and education 10%, with configurable normalized weights. Market uses up to five highest-scoring target-role vacancies; relevant experience is capped at 180 non-overlapping days. These weights have not been empirically validated as predictors of employment.

AI functionality: implemented baseline and optional learned extension

The implemented decision-support baseline is deterministic: it extracts known skill terms and aliases, combines profile evidence, scores vacancy requirements and derives gap-based actions. This is a knowledge-based matching pipeline, not a trained employment predictor or a generative assistant. The learned AI extension is a local sentence-embedding adapter; it is disabled in the verified configuration. The report therefore separates working rule-based functionality from available but unevaluated machine-learning functionality.

Capability	Actual mechanism and output	Evidence / current status
Skill extraction and normalization	Whole-term catalogue/alias matching in project descriptions, experience text and certificate names; canonical skill IDs and evidence sources are retained.	extract_skills() and competencies(); implemented and covered by tests. This is not an LLM or a general CV parser.
Proficiency-aware vacancy matching	Requirement weights and stored skill levels produce a coverage score and matched/partial/missing groups. Inferred evidence contributes at most 0.5; no requirements produce a null score.	score_requirements() and analyze(); implemented deterministic baseline, algorithm weighted-skills-1.0.
Career recommendations	Missing and partial requirements are sorted by importance and converted into skill-specific development actions. Readiness combines profile evidence with vacancy coverage.	analyze() and career_summary(); implemented rules, not learned generation or a probability of hiring.
Semantic skill matching	Optional SentenceTransformer encodes skill names; normalized dot-product similarity selects a candidate above the configured threshold. Semantic evidence remains partial.	semantic_candidate(); adapter exists, default threshold 0.85. Real model disabled/unavailable in the verified run; controlled tests and fallback only.
Learned ranking or employment prediction	A model trained on local hiring outcomes, validated success probabilities and an adaptive recommender are not implemented.	Future research possibilities; no effectiveness result is claimed.

Illustrative calculation, not a real student result: Python level 3 of required 3 with weight 2 contributes 2; SQL level 1 of required 2 with weight 2 contributes 1; absent Git with weight 1 contributes 0. The score is 100 × (2 + 1 + 0) / (2 + 2 + 1) = 60%. Python is matched, SQL is partial and Git is missing. This demonstrates an explainable calculation, not measured recommendation accuracy.

Implementation reference: https://github.com/vbl0-0/AITU-Career/blob/prob-period-1/backend/app/ai/matching.py. Matching correctness tests verify rules and boundaries. Only an independent relevance experiment can establish whether this approach ranks useful vacancies better than simpler alternatives.

7. Architecture, Diagrams and Data

Students/graduates and administrators use React, FastAPI and PostgreSQL. An explicit HeadHunter API adapter supports import, refresh, deduplication and source/availability tracking; live provider access was not available and no real import is claimed. University SSO and DU/Moodle are not integrated. Default keyword analysis needs no external AI service; the optional embedding model is disabled in the verified configuration.

System Context — AITU Career

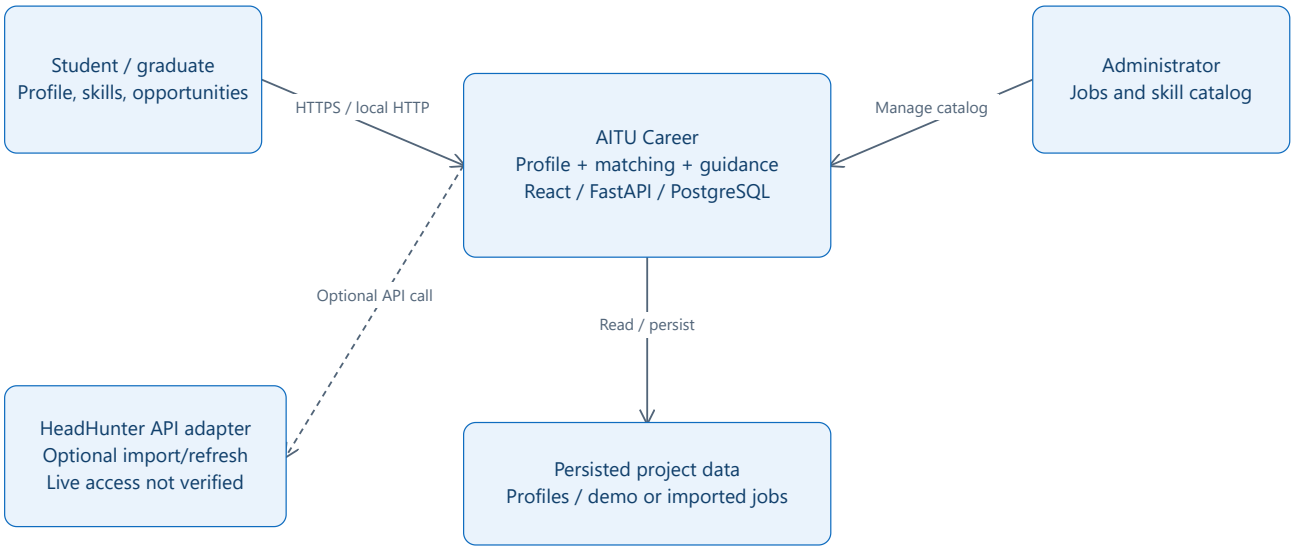


Figure 1. System context and responsibility boundaries.

Use Case Diagram — protected application scenarios

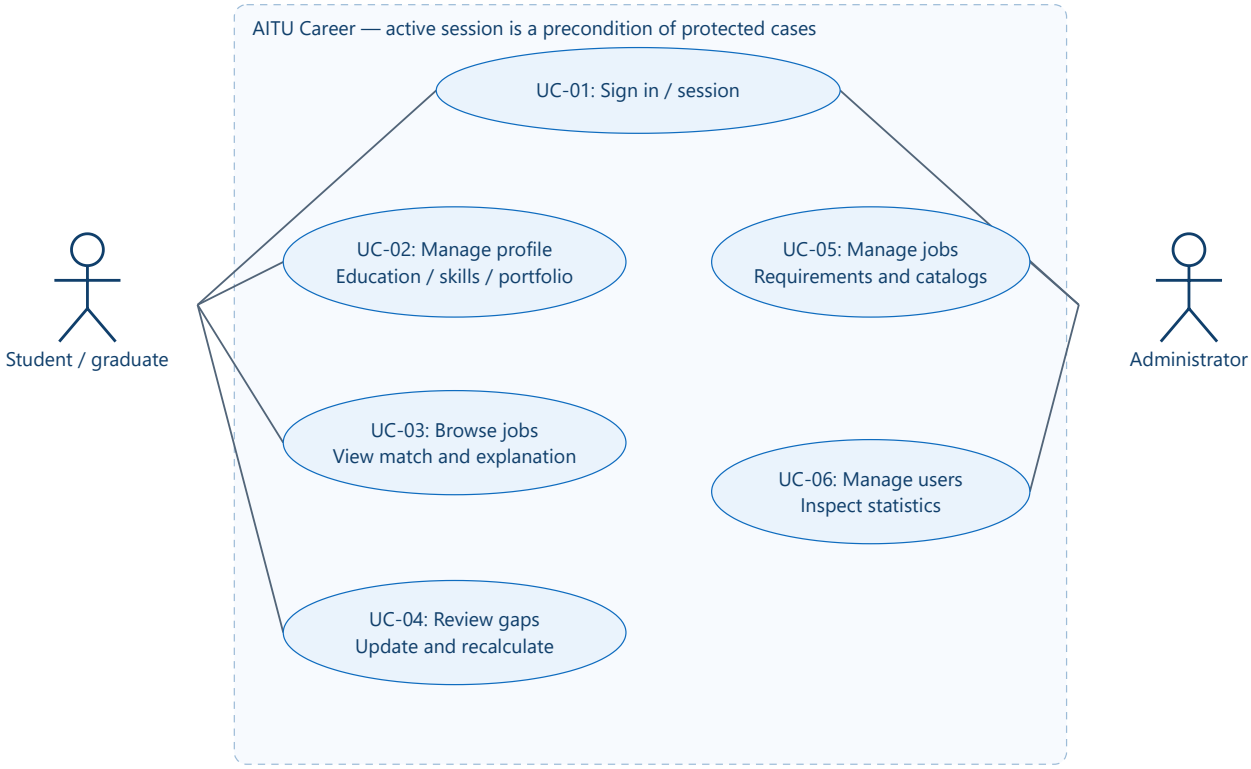


Figure 2. User roles and protected use cases.

Component Diagram — modular monolith

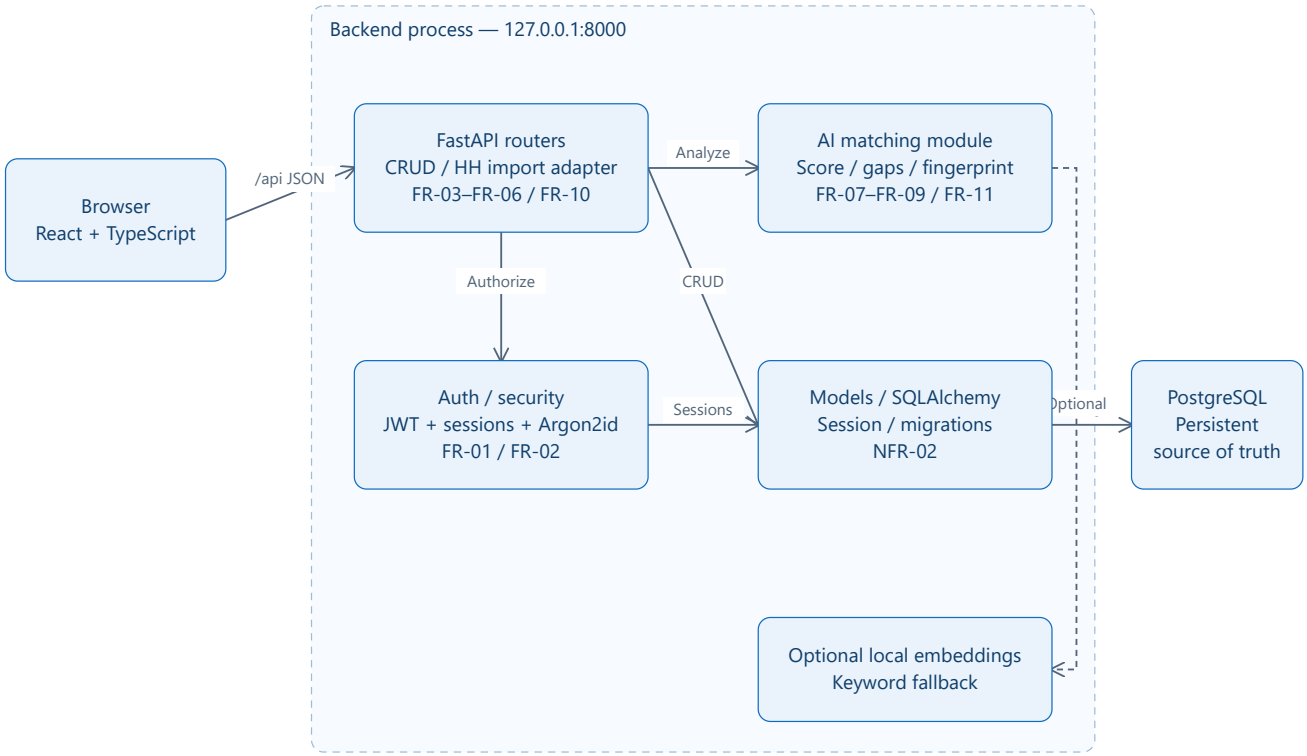


Figure 3. Draft component architecture aligned with the implementation.

Component	Responsibility	Related requirements
frontend/src/pages and components	Profile, vacancies, analysis and admin screens; routes and error states.	FR-03–FR-10; NFR-04/05
auth routes and core/security.py	Passwords, JWTs, sessions and permission checks.	FR-01/02; NFR-01
profile/jobs/admin routes	Input validation, CRUD, catalog and application journeys.	FR-03–FR-06; FR-10/12
ai/matching.py	Competencies, score, gaps, recommendations, fingerprint and optional semantics.	FR-07–FR-09/11; NFR-07
models/database/migrations	Relationships, constraints, persisted analyses and repeatable seed.	NFR-02/03

Data relationships: User 1:1 GraduateProfile; GraduateProfile 1:N Education, UserSkill, Project, Certificate, Experience, MatchResult and Recommendation. UserSkill/JobSkill link profiles/jobs to Skill; ProjectSkill links project technologies. SkillAlias stores normalization rules. CareerRole and categories are controlled dictionaries. User has AuthSession records; SavedJob links users to jobs. MatchResult preserves a calculated snapshot and its input/version context.

Analysis flow: authenticate and authorize → validate profile/job → normalize catalog terms → collect structured and textual evidence → obtain requirements → calculate weighted coverage → group matched/partial/missing requirements → derive development actions → persist the result with revisions → display it. Input changes do not silently replace the saved snapshot; they flag it as stale.

API route / resource	Purpose
POST /api/auth/register, /login, /refresh, /logout	Account creation and session management.
/api/profile, /api/skills, /api/projects, /api/certificates, /api/experience	Profile and portfolio operations; supported methods depend on the resource.
GET /api/jobs; GET /api/jobs/{id}; POST /api/jobs/{id}/match	Vacancy discovery, details and matching.
GET /api/recommendations; GET /api/admin/statistics	Career overview and administrator statistics.
GET /api/health; /docs	Health check and generated API contract.

The full current API contract is available locally at http://localhost:8000/docs. Diagram deliverables include editable SVG, JSON layout models, Mermaid/PlantUML sources and PNG exports. Diagram changes must remain consistent with the code and requirement IDs.

Editable diagram sources and exports: https://github.com/vbl0-0/AITU-Career/tree/prob-period-1/docs/pre-defense-1/diagrams. Local OpenAPI: http://localhost:8000/docs. The API is not publicly hosted by the static prototype viewer.

8. UX and Prototype

Public screen walkthrough: https://vbl0-0.github.io/AITU-Career/. It presents actual captures of the local application with navigation between screens, not a hosted backend or live editing service. The complete application runs locally using the published development instructions. The interface supports Kazakh, Russian and English; the report is in English. Original screenshot and diagram colors are preserved.

Journey	Steps	Expected outcome
UC-01: Start	Landing → registration/sign-in → dashboard.	An authenticated session and an accessible profile.
UC-02: Profile	Education program → desired role → skills → project.	Saved data and explicit confirmation.
UC-03: Match	Search → vacancy → score → detailed analysis.	Requirement groups and a traceable calculation.
UC-04: Improve	Add FastAPI → open an old analysis → recalculate.	A stale warning followed by a result based on current inputs.
UC-05: Administer	Admin sign-in → jobs → create/update.	Permission checks and validated skill requirements.

UX states include loading, no records, validation errors, server failures, save confirmation and stale results. Focus indication, labels and accessible action names are implemented; a comprehensive accessibility audit remains outstanding. A user-evaluation task asks participants to find a relevant vacancy, identify two gaps and explain the score.

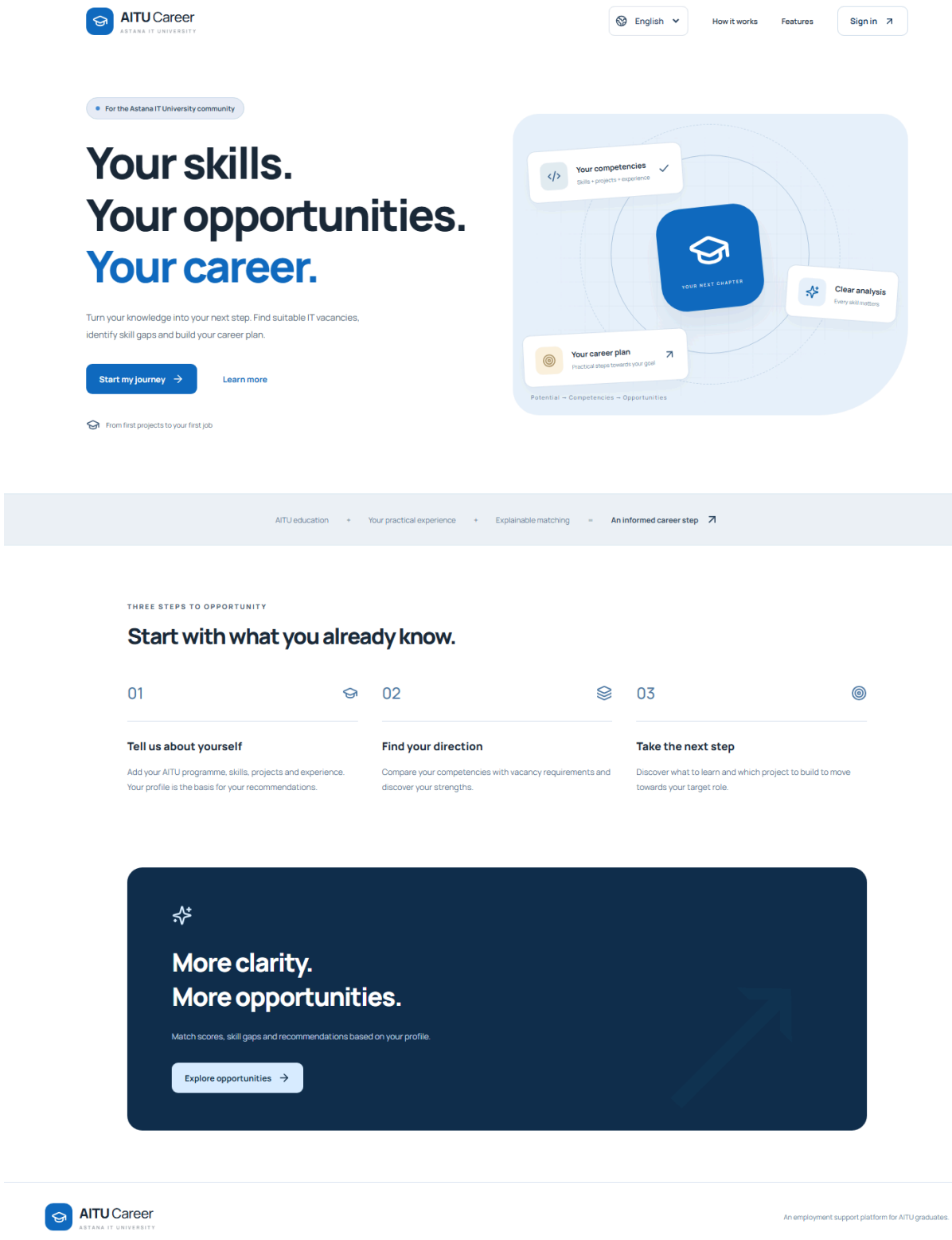


Figure 4. Entry to the graduate journey.

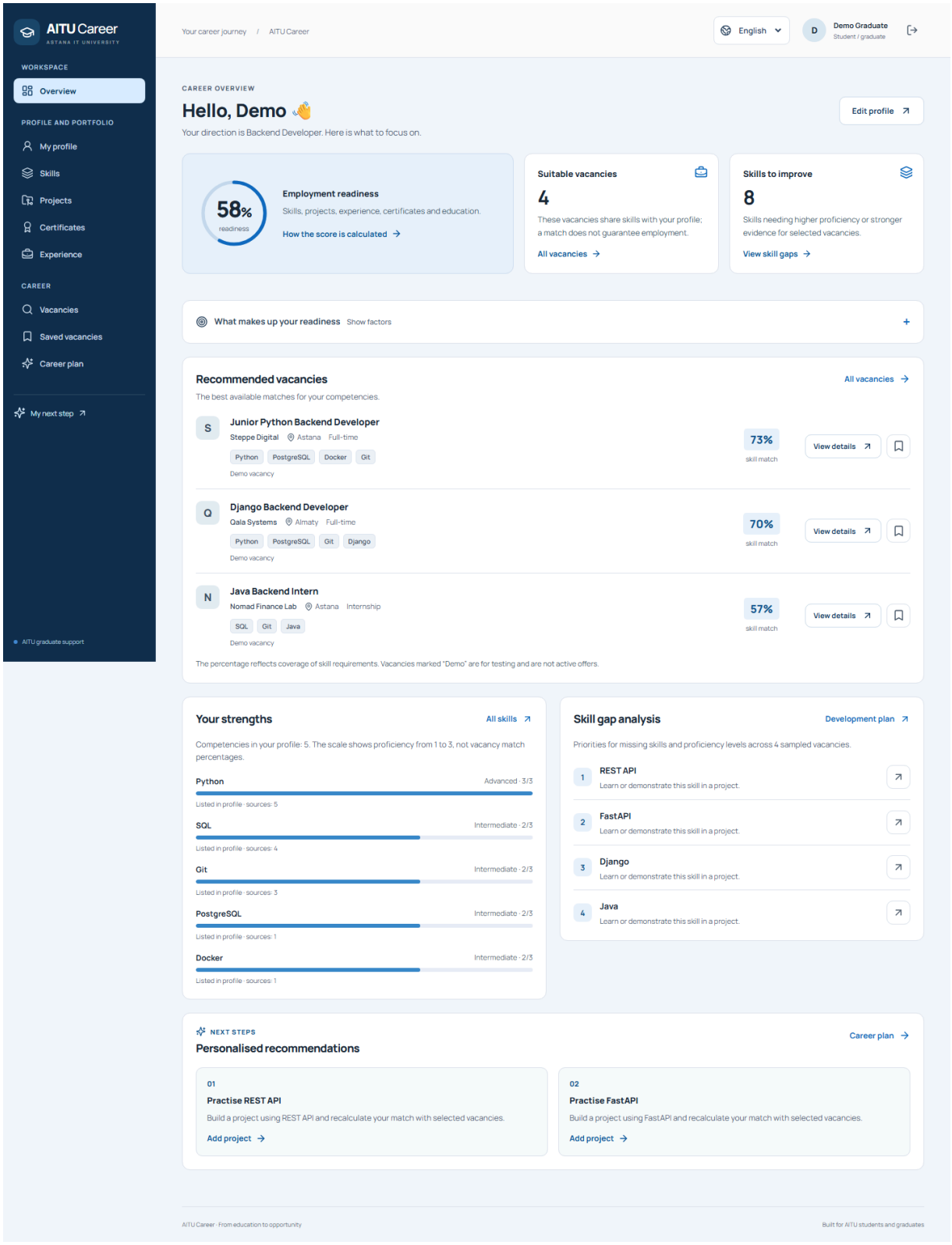


Figure 5. Demonstration graduate dashboard.

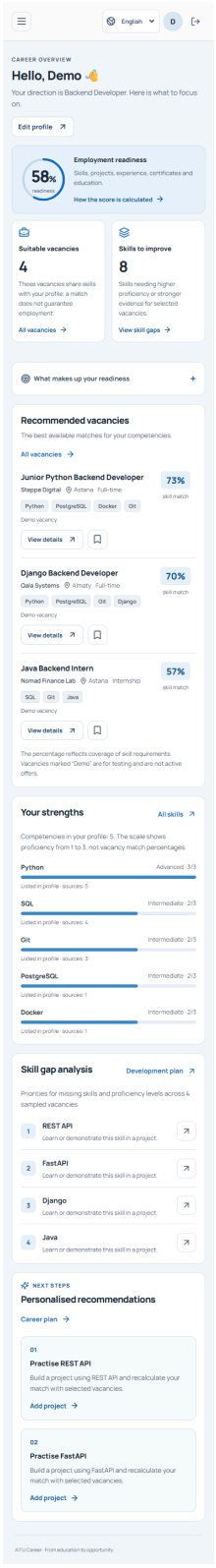


Figure 6. Mobile interface at 390×844.

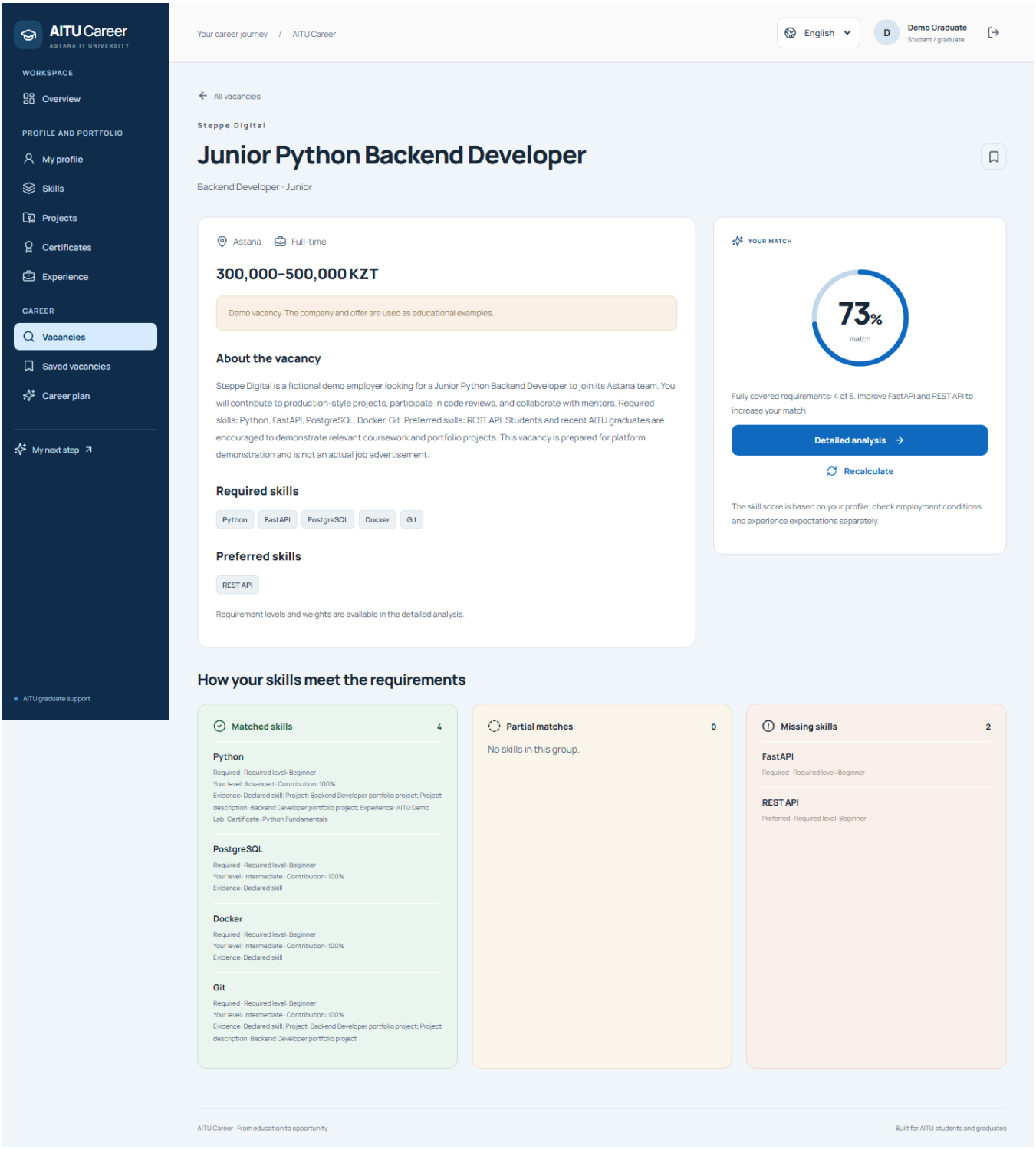


Figure 7. Vacancy requirements and explained matching.

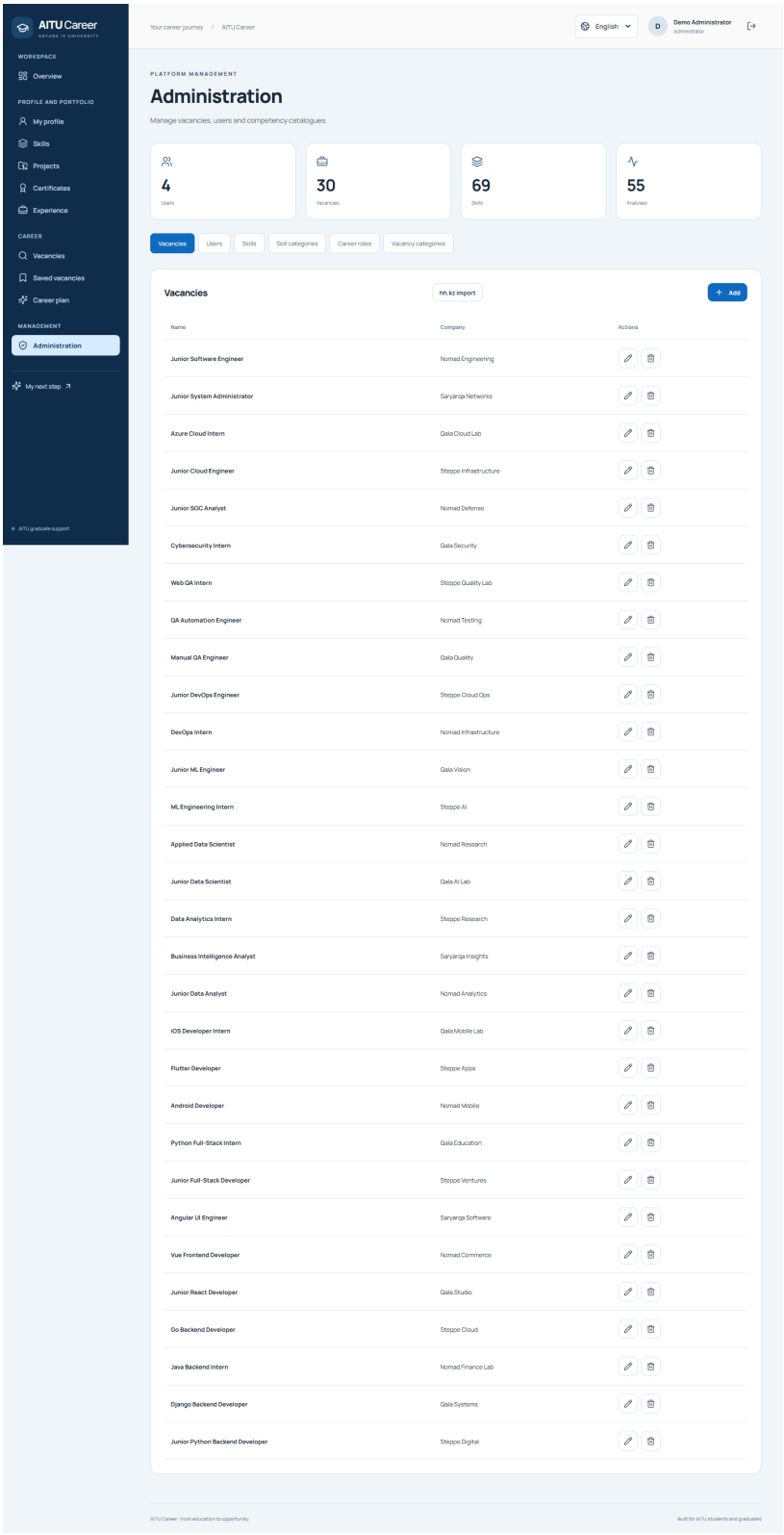


Figure 8. Administrative vacancy management.

9. Verification and Evaluation Plan

Verification checks implementation against requirements. Evaluation examines ranking quality and user understanding. Passing software tests does not demonstrate improved employment outcomes or model accuracy on the real labor market.

Verification on 24 October 2026	Observed result
python -m pytest	passed: 60 checks passed. 35.06 seconds; one Starlette TestClient/httpx deprecation warning. The initial database connection timeout was resolved by starting the existing project cluster.
npm run test:e2e	passed: 20 checks passed. Chromium: graduate and admin journeys, portfolio CRUD, Dashboard states, role restrictions, responsive navigation, kk/ru/en and vacancy import/salary regressions. Some focused tests use API fixtures; the core journey and multilingual pages also use the real local API.
npm run check:i18n	passed: 459 checks passed. Translation keys, interpolation parity and static references in kk/ru/en.
npm run build	passed: TypeScript and production Vite build. Existing main-chunk warning: 501.13 kB before gzip.
python -m alembic upgrade head; python -m alembic check	passed: Migrations applied; no new upgrade operations detected.
node --test .github/scripts/project-sync.test.mjs	passed: 9 checks passed. Reference parsing, trusted PR filtering, transitions, partial merges, pagination and status-only/idempotent mutations.
Targeted UI review (eleven routes, desktop/mobile)	passed: 22 checks passed. Screenshots inspected; no horizontal document overflow or blocking page alert at 1440px/390px. Keyboard skip-link focus has a 3px outline. This is not comprehensive WCAG conformance.

Machine-readable results: https://github.com/vbl0-0/AITU-Career/blob/prob-period-1/docs/pre-defense-1/evidence/validation.json. The initial connection timeout was resolved by starting the existing project PostgreSQL cluster; no application tables were replaced. Software checks do not establish ranking quality, real-market coverage or improved employment outcomes.

Level	Verification focus	Artifact
Unit	Formula, levels, boundaries, aliases, partial evidence and semantic fallback/cap.	backend/tests/test_matching.py
API and database	Sessions, roles, ownership, CRUD, constraints and staleness.	backend/tests/test_api.py
Browser	Graduate journey, portfolio, administration and mobile viewport.	frontend/tests/career.spec.ts
Structure	Type checking, production build and model/migration consistency.	npm run build; alembic check

Level	Verification focus	Artifact
Quality evaluation	Independent labels and a fixed baseline comparison.	Protocol below; experiment not yet performed

Operational evaluation protocol in response to SUP-01 and SUP-02

The following is a next-stage protocol, not a completed experiment. Freeze 20 anonymized or synthetic profiles and 50 legally usable vacancies, using at least five candidates per evaluated profile. Use ten profiles for development and ten held-out profiles for testing; store IDs, source permissions, split membership and data hashes. This produces 500 development and 500 held-out profile–vacancy pairs. Where development and test share vacancies, the stated claim concerns new profiles rather than unseen vacancies.

Comparison	Controlled change	Result status
B0 — exact overlap	Case-normalized canonical skill-name overlap, without aliases, proficiency levels or requirement weighting.	Not yet measured.
B1 — current baseline	Current catalogue/aliases, evidence levels and requirement weights, with semantic inference disabled.	Implementation verified; ranking quality not yet measured.
B2 — semantic extension	B1 plus the optional local embedding adapter, with model revision and threshold fixed before the held-out run.	Real model not evaluated. If it cannot be loaded, record B2 as not run; do not substitute a fabricated score.

Two raters independent of the matching implementation should assign relevance grades 0 (irrelevant), 1 (partly relevant) and 2 (suitable), without seeing algorithm scores. Store both original ratings and weighted Cohen's kappa before adjudication. Rater identities and actual dates remain unfilled until the work occurs. Tune aliases and semantic thresholds only on development data; use the same eligible candidates for all methods and break score ties by vacancy ID.

Measure	Calculation / procedure	Decision and retained artifact
Ranking relevance	Precision@5 = count of grade-2 vacancies in the first five / 5. nDCG@5 uses gains 2^grade − 1 and log2(rank + 1) discount, normalized by the ideal top-five ranking.	Macro-average per held-out profile. Profiles with zero ideal gain have undefined nDCG and are counted separately; report their number and candidate coverage. Preserve per-profile rankings and labels.
Extraction accuracy	Micro precision, recall and F1 against annotated canonical skill mentions; show separate Kazakh/Russian/English results and representative errors.	Keep span/skill reference annotations and the exact normalization rules. No extraction-accuracy figure is currently claimed.
Efficiency	Same CPU, RAM, data, process and cache conditions; 30 warm-up requests, then 200 measured requests. Record median and p95, model loading separately, and errors.	Keyword p95 target is at most 500 ms. Pre-register a separate semantic latency/memory budget before comparing B2. Preserve request timings and environment details.

Measure	Calculation / procedure	Decision and retained artifact
Uncertainty and adoption	Use seed 42 and 1,000 paired profile-level bootstrap resamples for the nDCG difference; report a 95% interval and Precision@5 changes.	Adopt an extension only if held-out nDCG improves with an interval above zero, Precision@5 does not deteriorate, its registered resource budget is met and functional regressions are absent. Otherwise retain the simpler method and report the negative outcome.

Execution responsibilities are planned: Zhan coordinates the dataset, annotation records and error analysis; Ayadil prepares the reproducible B0/B1/B2 runner, model/configuration hashes and latency measurements. Both inspect the held-out results and review each other's experiment changes. Independent raters must be recruited; account ownership or an AI-assisted check is not treated as expert relevance annotation. The tracker schedules label preparation in #10 and baseline/latency evaluation in #11 for the next stage.

Expected evaluation artifacts: a dataset/split manifest, independent labels and agreement record, a committed comparison runner, environment/model configuration, per-profile rankings, metrics/timing files and an interpretation of errors and negative results. These artifacts are planned and are not listed as already available. The 60 backend and 20 browser tests establish implementation behavior, not AI effectiveness.

Validity threats include convenience sampling, self-reported skills, synthetic vacancies, language bias, tuning leakage, facilitator effects and a small profile sample. Mitigations are independent annotation, a held-out split, negative examples, fixed comparison conditions and retention of raw judgments and errors. Docker and the real embedding model were not exercised in this environment; controlled tests do not replace their evaluation.

10. Project Management, Responsibilities and Risks

Responsibilities below are supported by dated account-linked delivery records. OpenAI Codex assisted implementation, packaging, technical verification and publication. These records establish what was delivered under each account; imported history alone does not establish exclusive original authorship or the time each person spent. Actual human peer review remains to be added in GitHub, as requested by the supervisor.

Participant / work area	Concrete delivered artifacts and traceability	Evidence limit
Қожабек Аяділ Нұрланұлы / @vbl0-0	Backend API/models/migrations: 6c434f2; regression tests/container setup: 0b8c8bb; architecture/startup delivery: 26c8605. Related PRs #19, #21, #23 and issues #7/#9/#12/#14.	AI-assisted current deliveries. No percentage, independent initial code authorship or unrecorded manual review is inferred.
Амантай Жан Әділұлы / @igiiaw	Frontend/i18n/browser tests: 7717652; references/documentation: c12021f; report sources: 203f7c7; report/evidence exports: d308bb1 and adc5d46. Related PRs #20/#22/#24/#25 and issues #4/#5/#8/#16/#17.	AI-assisted current deliveries. Account attribution and module responsibility are distinguished from proof of who wrote every imported line.

Merged delivery records: https://github.com/vbl0-0/AITU-Career/pull/19 (repository automation); https://github.com/vbl0-0/AITU-Career/pull/20 (frontend); https://github.com/vbl0-0/AITU-Career/pull/21 (backend and regression tests); https://github.com/vbl0-0/AITU-Career/pull/22 (research and report sources). Further delivery PRs are recorded in the repository and linked issue comments. Original commits are preserved through merge commits.

Tracker: https://github.com/users/vbl0-0/projects/1. The 18 original tasks retain their source IDs, priorities, planned owners and original deadlines. GitHub Issues contain actual acceptance criteria, account assignments and completion evidence. Optional interview/feedback tasks are distinguished from completed work. Later-stage independent labels and matching-quality experiments are not marked complete.

Risk or remaining limit	Response
Static prototype versus the complete application	The public viewer is clearly labelled; use the development guide to run authenticated CRUD and matching.
Demonstration vacancies mistaken for live jobs	The seed dataset is fictional and labelled. No live HeadHunter import or current market representativeness is claimed.
Unmeasured model or employment outcomes	Keep the evaluation protocol; do not infer efficacy from passing implementation tests.
Confusing account history with individual authorship	Preserve real dates and links; state the use of AI assistance and distinguish delivery from historical authorship.
Documents differ from the submitted version	Use the exact prob-period-1 tag and the matching PDF exports; do not move the submitted tag.

Risk or remaining limit	Response
Optional evidence shown as mandatory	Record the team clarification and keep future interviews/reviews optional.

Submission version: https://github.com/vbl0-0/AITU-Career/tree/prob-period-1. One representative submits both students' materials before 18:00 UTC+5 on 24 October 2026. A form confirmation is retained only after actual submission.

Tracker snapshot: 14 completed, 1 optional activities closed as not planned, and 3 open. GitHub's built-in closed-item workflow places both closure types in Done; the issue state reason distinguishes cancelled scope from performed work. Snapshot: https://github.com/vbl0-0/AITU-Career/blob/prob-period-1/docs/pre-defense-1/evidence/github-current-issues.json.

Development history supported by original Git metadata

The original Git metadata records repository setup, automation and frontend import on 23 October 2026, followed by backend, report-source, architecture and submission deliveries on 24 October 2026 (UTC+5 author timestamps). These are the real recorded publication stages; no earlier incremental history is reconstructed. The pre-feedback snapshot contains 11 non-merge commits, preserved with their original author dates and subjects: https://github.com/vbl0-0/AITU-Career/blob/prob-period-1/docs/pre-defense-1/evidence/development-history.json.

Peer review process to introduce in GitHub

The checked PRs #19–25 contain no submitted GitHub review records. Their merge status, automated tests and supervisor feedback are not peer approval. This remains an improvement action, not a completed feature. For the next substantive backend change, Zhan should review Ayadil's PR; for a frontend/report change, Ayadil should review Zhan's PR. Each reviewer personally inspects the diff, runs the relevant checks and records concrete comments and test evidence. The author links resulting fixes, and the reviewer rechecks before approval. Only actual actions may be recorded under a person's account.

Use small task-focused PRs, a contribution log stating the participant's actual implementation/test/review actions and any AI assistance, and a checklist covering correctness, authorization/data handling, edge cases and regression evidence. Where repository settings support it, require one independent approval and passing relevant checks before merging; enabling that rule is planned, not claimed as already configured. Preserve review comments and resulting commits rather than creating approvals after the fact. Peer-review status evidence: https://github.com/vbl0-0/AITU-Career/blob/prob-period-1/docs/pre-defense-1/evidence/peer-review-status-2026-10-24.json.

11. Supervisor Feedback and Response to Comments

Supervisor feedback was supplied to the team on 24 October 2026. The following record reproduces the two messages transcribed by Ayadil, attributed to Оразова Арайлым Жанболаткызы (display name Arailym). The original private chat was not independently accessed and the messages do not identify an exact reviewed commit. Their positive assessment is recorded without turning it into an unconditional final approval.

SUP-01 — 24 October 2026, 15:28; original English message

“The team has made good progress in project analysis, system design, documentation, and initial implementation. The report, GitHub repository, task tracker, and prototype are well organized and generally meet the objectives of the current stage. For the next stage, the team should focus on further system testing, evaluating the matching approach, and improving development and code review practices.”

SUP-02 — 24 October 2026, 15:29; English translation of the Russian message

“Overall, the team has worked well and the report is fairly complete. However, several aspects need improvement: make the AI functionality and the method for evaluating its effectiveness clearer. It is also important to show each participant's actual contribution, the development history, and to add peer review in GitHub.”

The original Russian text, English translation, times and source limitations are retained in https://github.com/vbl0-0/AITU-Career/blob/prob-period-1/docs/pre-defense-1/evidence/supervisor-feedback-2026-10-24.json.

The previous prepared snapshot is retained at https://github.com/vbl0-0/AITU-Career/tree/prob-period-1-before-feedback-20261024. It is a preserved pre-feedback assembly, not a claim that the supervisor reviewed that exact commit. The current prob-period-1 package incorporates the reporting changes below before form submission.

Supervisor comment	Change in this report	Implementation / next-stage status
Clarify the AI functionality.	Section 6 now distinguishes deterministic extraction/matching/recommendation rules, the optional embedding adapter and functionality that does not exist; includes a worked calculation and code reference.	Documentation updated. No new trained model or generative service is claimed.
Explain how matching effectiveness will be evaluated.	Section 9 specifies B0/B1/B2, data splits, blind independent labels, metrics, latency conditions, adoption criteria and retained artifacts.	Protocol clarified. Independent quality and latency experiments remain planned in #10/#11.
Show each participant's actual contribution and development history.	Section 10 lists concrete account-linked artifacts, commit IDs, PRs, real author dates and limits of historical authorship attribution.	Published evidence documented. Subsequent contribution logs must record actual personal implementation, testing and review actions.

Supervisor comment	Change in this report	Implementation / next-stage status
Add peer review and improve development practices.	Section 10 defines cross-review responsibilities, task-focused PRs, review evidence, follow-up fixes and a proposed independent-approval gate.	Not yet performed/configured. No GitHub peer approval is fabricated.
Continue system testing.	Section 9 preserves actual 60-backend/20-browser verification separately from independent model evaluation and identifies untested areas.	Current verification recorded; further regression coverage and quality experiments belong to the next stage.

Documentation changes above address the current reporting comments. The remaining engineering and evaluation actions are explicitly prospective. Receiving supervisor feedback does not substitute for independent GitHub peer review, independent relevance labels or measured matching effectiveness.

Appendix A. References

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Appendix B. Submission Checklist and Evidence Status

Submission form: https://forms.cloud.microsoft/r/Yy6AXT5pe3. The provided screenshots specify a repository URL, the exact prob-period-1 version URL, individual contributions, current status, one board/task PDF (maximum 10 MB) and one consolidated report PDF (maximum 100 MB). This report contains the eleven main sections in order and a table of contents.

Requirement	Prepared artifact	Scope or remaining limit
Repository and version	https://github.com/vbl0-0/AITU-Career; https://github.com/vbl0-0/AITU-Career/tree/prob-period-1	The tag identifies the submitted package.
Identity and contributions	Confirmed names, groups, supervisor and account-linked delivery records.	AI assistance and current-delivery limits are disclosed.
Research and comparisons	17 sources including eight scientific publications; four systems and eight common criteria; five technology decisions.	Local user demand and real matching quality are not empirically established.
Requirements and architecture	Testable FR/NFR criteria and editable diagram sources/exports.	https://github.com/vbl0-0/AITU-Career/tree/prob-period-1/docs/pre-defense-1/diagrams
Prototype	https://vbl0-0.github.io/AITU-Career/	A public screen walkthrough; the complete API-backed application runs locally.
Verification	Current backend, browser, build, translation and migration results.	Docker runtime and real embedding-model evaluation were not performed.
Interviews and supervisor feedback	No interviews; actual supervisor messages of 24 October and the response matrix are recorded in Section 11.	Evaluation and human peer review remain next-stage work.
Task evidence PDF	tracker-evidence.pdf with genuine board and task captures.	Tasks show actual progress; later-stage work can remain open.
Unified editable report	report.docx, report.html and matching report.pdf.	Use the final named PDF copy for the form.
Form submission	Prepared answers and both upload files.	The representative must verify the final form page and retain the actual confirmation.

Report upload filename: SE-2425_Kozhabek_ProbPeriod1.pdf. Deadline: 24 October 2026 before 18:00 UTC+5. Known fields are completed; any genuinely unknown personal information stays blank. The package does not assert that the form has already been submitted.